

Jiguang Li

Ph.D. Candidate in Econometrics and Statistics, University of Chicago Booth School of Business

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Education

University of Chicago Booth School of Business

Chicago, IL

Ph.D. in Econometrics and Statistics, expected June 2027; M.B.A., expected June 2027

2022–2027

Advisor: Veronika Ročková

Research interests: Bayesian statistics, sequential decision-making, psychometrics, causal inference.

Yale University

New Haven, CT

M.A. in Statistics

2019–2020

Middlebury College

Middlebury, VT

B.A. in Mathematics and Computer Science

2015–2019

Summa Cum Laude; Highest Honors in Mathematics; Davis UWC Scholar.

Publications and Preprints

Jiguang Li, Sid Kankanala, and Veronika Ročková. “Empirical Likelihood with Generative AI.” *Journal of the American Statistical Association* (R&R), 2026. [\[preprint\]](#) [\[code\]](#)

Jiguang Li and Hengrui Luo. “Robust Bayesian Optimization via Tempered Posteriors.” *Journal of the Royal Statistical Society: Series B* (R&R), 2026. [\[preprint\]](#) [\[code\]](#)

Bengusu Nar, **Jiguang Li**, Veronika Ročková, and Panos Toulis. “Dynamic Treatment on Networks.” *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. [\[preprint\]](#)

Jiguang Li, Robert Gibbons, and Veronika Ročková. “Deep Computerized Adaptive Testing.” *Psychometrika*, 91(3): 1127–1148, 2026. [\[paper\]](#) [\[code\]](#)

Jiguang Li, Robert Gibbons, and Veronika Ročková. “Sparse Bayesian Multidimensional Item Response Theory.” *Journal of the American Statistical Association*, 120(552): 2592–2605, 2025. [\[paper\]](#) [\[code\]](#)

Core Skills

Methods: Bayesian modeling, causal inference, sequential decision-making, reinforcement learning, optimization, latent-variable models.

Programming: Python (PyTorch, PyMC), R, Java.

Languages: Chinese (native), English (fluent).

Experience

Applied Scientist Intern

Amazon Web Services

Marketing Data Science & Engineering

Jun–Sep 2026

- Developed causal targeting methods for AWS executive account prioritization using heterogeneous treatment effects estimated from longitudinal panel data.

Applied Scientist Intern

Amazon

Grocery Economics & Optimization Science

Jun–Aug 2025

- Developed Bayesian marketing-mix models and a constrained optimization framework for media-channel budget allocation for Whole Foods Market and Amazon Fresh.

Full-Time Research Professional, Center for Applied AI

University of Chicago

Supervised by Sendhil Mullainathan

Aug 2020–Jul 2022

- Built scalable PyTorch pipelines for medical imaging, including CNNs, meta-learning, multi-GPU training, and StyleGAN2.

- Developed Bayesian inference workflows in PyMC/HMC for educational testing and analyzed large-scale item-response, radiograph, and criminal-justice datasets.

Summer Research Assistant in Astrostatistics

California Institute of Technology

Supervised by George Djorgovski and Eilat Glikman

Summers 2017–2019

- Built a quasar variability analysis pipeline and implemented two-sample testing procedures; received the Caltech VURP Award in 2017.

Service and Teaching

Journal Reviewer: *Bayesian Analysis, Operations Research, Psychometrika, Statistics and Computing.*

Conference Reviewer: NeurIPS 2026.

Teaching Assistant: Business Statistics (MBA), Chicago Booth; Linear Algebra and Macro Theory, Middlebury College.

Selected Presentations

Seminar, Chicago Booth Econometrics & Statistics Ph.D. Seminar, Chicago, IL, December 2026.

Seminar, Committee on Quantitative Methods Workshops, University of Chicago, Chicago, IL, October 2026.

Invited talk, ICSA Applied Statistics Symposium, Arlington, VA, June 2026.

Seminar, Chicago Booth Econometrics & Statistics Ph.D. Seminar, Chicago, IL, October 2025.

Seminar, Department of Public Health Sciences, University of Chicago, Chicago, IL, October 2025.

Invited talk, Chicago Booth Healthcare Initiative, Chicago, IL, November 2024.

Contributed poster, ISBA World Meeting, Venice, Italy, July 2024.

Contributed talk, Junior ISBA Conference, Venice, Italy, June 2024.

Selected Coursework

Statistics/ML:	Bayesian Deep Learning, Causal Machine Learning, High-Dimensional Hypothesis Testing, Mathematical Statistics.
Probability/Math:	Advanced Probability, Measure Theory, Optimization Techniques, Spectral Graph Theory.
Economics:	Bayesian Econometrics, Empirical Analysis, Price Theory.
MBA:	Marketing Data Science, Investments, Operations Management, Financial Accounting.